

Chapter 2

This chapter describes the process through which simple but previously hidden 2D and 3D geometric structures were uncovered.

At the outset, nothing certain was known about how dipole function might be embedded in the violin's geometry.

I began by testing a simple idea, and through more than twenty different attempts at geometric construction — a process that required three months of continuous work — I gradually uncovered conditions that produced functional results.

Each iteration was drawn manually, point by point, often using hundreds of reference points, and every iso-line layout was calculated and examined in detail.

These drawings revealed whether the construction produced, or failed to produce, the dipole function I sought.

The decisive problem was how to shape the baseline — the only variable within the 2D geometric layout.

The baseline binds together the structures of the upper and lower bouts. Its shape determines the constructional layout of the iso-lines, and only when correctly formed does it produce the condition necessary for dipole function.

2.1 How the Research Process Generated the Specific Iso-Line Layouts

What I sought was a solution in which any cross-section drawn from the outline toward the center of the **upper bout structure** would always have an exact counterpart in the **lower bout structure**.

This congruence is the defining mark of dipole condition. In other words, the geometry had to ensure structural equivalence between the two regions, even though their widths differ.

To test this, I calculated iso-lines across the arching layout. On each iso-line, a point was marked at **5 mm internal distance**.

This painstaking work required repeating the process across many layouts, each time checking whether the construction achieved the condition I was searching for.

The effort was demanding, but it gradually revealed how dipole behaviour could be **embedded in the violin's form through geometric congruence**.

Each failed attempt clarified the limits of the baseline's shaping, while each successful iteration confirmed that the dipole condition could be constructed — not assumed.

Here it will become explained how a simple geometric construction reveals the conditions under which dipole behaviour emerges.

The conditions described here not only produce a breathing dynamic but also give rise to constructive interference within the corpus, together with a structural framework that will be explained in Chapter 3.

There, the arching system and its internal geometry will be shown to stabilize these functions, allowing the luthier to harness them more effectively in practice.

A holistic understanding of the violin's structural conditions is essential for defining its functional behaviour.

The goal is to develop a geometry-based framework that accounts for the diverse mechanical properties embedded within the instrument's arched contours.

Crucially, this design must incorporate dipole qualities — those that enable coordinated response across the corpus, even when asymmetries exist between bout structures.

Such an approach ensures that the instrument maintains dynamic equilibrium and structural coherence, allowing its internal spring-like behaviour to operate symmetrically around the sound post.

By leveraging geometric logic, the model supports consistent force transmission and deformation behaviour, regardless of asymmetries in width or curvature.

This implies that symmetrical integrity must be preserved on both sides of the sound post to sustain effective structural conditions — conditions that enable frequency-dependent resonance interactions to emerge and propagate across the body.

Looking ahead, Chapter 3 will build upon this foundation by presenting a specialized internal framework that significantly affects the function of the instrument.

It will show how the arching structure manifests both dipole and interference behaviour, and how this geometric logic can be harnessed by the luthier to achieve more consistent and refined results.

2.2 Structural Question and Model

This raises an important question: **Can a structural model be designed in which equal cross-arc shapes extend from the instrument's outline toward its center on both sides of the sound post — effectively achieving functional symmetry despite the inherent differences in bout geometry?**

Such a model would represent a breakthrough in violin design and in the understanding of its function. It would offer a unified structural framework that balances asymmetrical contours through geometric logic, enabling:

- Consistent stress distribution
- Coordinated deformation
- Enhanced dynamic responsiveness across the entire body

This design would not only unify the instrument's structural behaviour but also elevate its mechanical potential by leveraging geometry-driven reinforcement strategies.

The figure below illustrates how equal constructional conditions can be maintained within the arching shape, despite the differing widths of the upper and lower bouts.

Through precise geometric alignment, identical cross-arc shapes are distributed symmetrically across both plates — confirming consistency in structural logic and reinforcing the principle of coordinated deformation.

Here, we observe the emergence of a possible dipole function. The resulting configuration is shown in Figure 2.1 on the top plate (belly), and separately on the back plate. When equal structural composition is subjected to equal cross-sectional action, the structures may come into action, even if the areas are different. Dipole behavior may appear.

This was the underlying constructional idea we sought an answer to. Here the result is shown, but how that condition became possible through geometric construction is described later on.

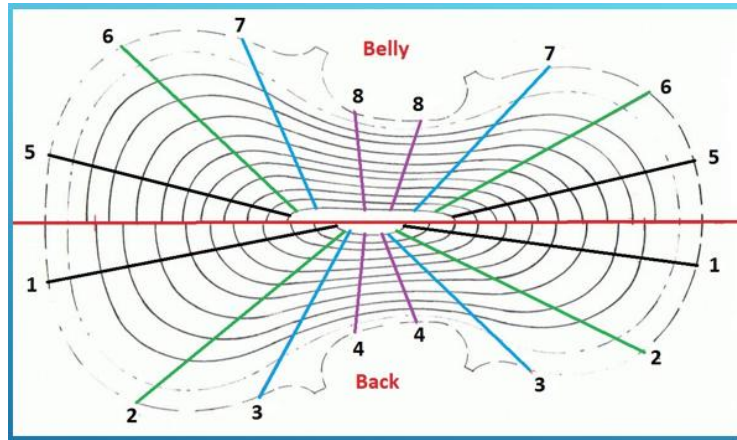


Figure 2.1

The top and back plate iso-line layout produces equal cross sections, indicated by equally coloured cross sections.

The iso-lines are calculated at 5 mm internal spacing from the outline toward the center.

Each colour represents a congruent cross section on the upper and lower bout areas, confirming that equal geometric conditions can be maintained despite differences in width.

The solution produces thousands of such equal cross sections, so they are merely copies of each other.

This visual alignment demonstrates the principle of coordinated deformation that becomes produced shown in figure1.1.

Dipole function emerges.

2.3 Symmetry Within Cross-Arc Geometry and the Resulting Structural Dipole Quality

Figure 2.1 demonstrates that equal constructional conditions can be preserved within the arching, even though the upper and lower bouts differ in width.

A simple way to visualize this principle is through the metaphor of *reshaping a cake*.

Imagine drawing a diagonal line across a cake, cutting out a small triangular slice, rotating it, and closing the gap by inserting the removed piece.

The cake now has a different width and a slightly altered shape.

If this operation is repeated thousands of times, the result is a continuous family of shapes with varying widths between the removed and reinserted pieces.

This intuitive idea suggests a possible method by which the violin's form could have been constructed.

If such a transformation can be expressed geometrically, then the original maker may indeed have used a remarkably simple and powerful process—one capable of producing both structural coherence and acoustic quality.

Figure 2.2 and the Practical Search for a Geometric Solution

The central question, then, is how to formalize this intuitive “cake-reshaping” idea as a constructional concept grounded in geometry—rigorous, yet simple.

Only measured iso-line drawings of the arching have survived; the outline itself varies widely among historical instruments and therefore cannot serve as a reliable starting point for identifying a geometric structure.

Any valid method must rely on internal geometric logic rather than external contour templates.

The solution developed in this chapter provides exactly that.

By identifying the underlying geometric process, the complete structure of the violin becomes visible.

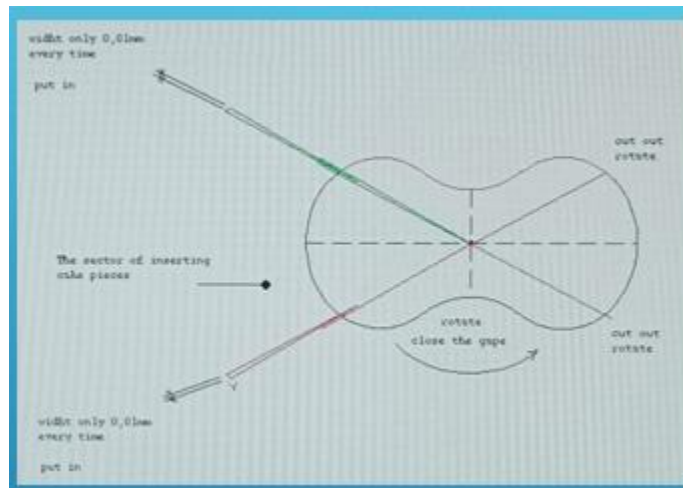


Figure 2.2

The solution found is what this chapter describes, and through it the complete structure of the violin became revealed.

Through **precise geometric alignment**, **identical cross-arc shapes** are distributed **symmetrically across both plates**—confirming **consistency in structural logic** and reinforcing the principle of **coordinated deformation**.

This symmetry suggests the presence of a dipole function—a dynamic behavior where opposing zones of the arching respond in complementary motion, channeling energy through balanced deformation.

The result is a structure that not only resists collapse, but actively breathes, flexes, and amplifies mechanical output through geometrically tuned interference conditions.

All of this remains theoretical, of course, because wood is a natural material and its exact response cannot be predicted.

We must assume that final corrections must be made on the finished instrument as it finds its optimal functional state.

How these structural conditions were constructed is detailed in the following sections.

2.4 How the Dipole Condition Was Constructed Using Baseline Geometry

A balanced opposition between the upper and lower bout structures is not naturally present in all instruments; it must be *constructed* through precise geometric alignment.

Figure 2.1 illustrates the outcome when this alignment is constructed upon a geometrically defined baseline—the abutment of all extended cross-sections—arranged such that, after calculation, the iso-line layout reveals the constructive dipole condition.

This baseline is derived from the extended cross-arc radii at rib height, which together establish a geometric axis governing the underlying three-dimensional structure of the iso-lines.

In effect, the abutment created by the extension of all cross--arcs establishes a possible geometric outcome.

When this baseline is combined with the **fixed-length arc shape**--a radial form--it enables the controlled placement of iso-lines.

The shape of the baseline indicates whether the arching system is capable of supporting dipole behaviour.

In the configuration shown, the baseline generates symmetrical iso-line contours across both plates, thereby confirming the emergence of a dipole structure.

Such an outcome is not the result of empirical averaging, inherited tradition, or chance; it derives from intentional geometric design.

By demonstrating that dipole behaviour can be embedded through baseline geometry, this construction provides a reproducible framework — a departure from centuries of empirical observation and a design logic that can be applied consistently by the luthier.

2.5 Solving the Constructional Challenge Through Geometric Precision

This design outcome would not only unify the instrument's structural behavior but also enhance its **dynamic responsiveness** and **mechanical potential** by leveraging **geometry-driven reinforcement strategies**.

This was the central challenge in solving the constructional problem—and the primary focus of the research: to identify a **constructional quality** capable of unifying structure and function through **geometric precision**.

Thickness regulations may influence the construction process and this will be explored in more detail later.

By integrating **geometric principles** with advanced techniques for **load distribution**, it becomes feasible to develop a design that **reconciles inherent structural disparities** within the violin body.

2.6 Dynamic Behaviour Through Structural Precision

Such a model not only ensures mechanical stability but also enhances the instrument's dynamic behaviour and its capacity for responsive transformation under playing conditions.

The illustrated cross-sections above demonstrate a uniform deflection potential, which facilitates the seamless operation of spring-like, volume-shifting functions as outlined in chapter 1.

This behaviour reflects the coordinated interaction between curvature, stiffness gradients, and internal force pathways, forming the basis for a structurally unified and dynamically expressive instrument.

The stiffness of the violin's structure — heavily influenced by the composition and thickness of the wood — is intricately tied to a precisely formed arching design.

Because the upper and lower bout structures differ in area, they may require varying plate thicknesses to achieve dynamically equivalent behaviour.

A more detailed exploration of this principle will follow in a later chapter focused on improvements in dynamic responsiveness.

Balancing these variables is essential for achieving optimal structural performance, which contributes significantly to the instrument's acoustic excellence.

This relationship is vital: the precise arching geometry not only defines the structural integrity of the violin but also governs the interplay between flexibility and rigidity.

By applying a geometry-based model, it becomes possible to calculate conditions previously unexplored yet structurally attainable — enhancing both quality and performance.

Such modelling enables a deeper understanding of structural dynamics and interaction effects, allowing for the fine-tuning of key parameters to reach exceptional functionality and expressive responsiveness.

2.7 Sacconi's Legacy and the Geometry of Arching

It became clear to me that Simone F. Sacconi (1895–1973) identified critical structural conditions in the arching shape during his tenure at the Rembert Wurlitzer workshop in New York.

There, he had the rare opportunity to examine a wide array of instruments crafted by Antonio Stradivari.

Sacconi likely discerned recurring geometric patterns in the arching and attempted to replicate these structures through empirical reconstruction.

However, we must recognize that Sacconi was studying violin structures that had been reshaped by **creep** over more than two to three centuries.

It is impossible to recover the original arching structures from such altered forms. Sacconi himself understood this limitation.

Figure 2.3 is shown what was found: Sacconi's construction of the transition from lower to upper-bout structure.

This was achieved by drawing angular lines and bisecting them, thereby predicting the location of iso-lines across the arching surface.

The configuration shown likely represents an averaged arching profile distilled from multiple Stradivari violins.

It reflects Sacconi's effort to extract a consistent geometric logic from empirical observation — a synthesis of craft intuition and structural analysis.

Yet, because his reconstructions were based on altered instruments, they could not reveal the original constructional framework embedded in the violin's design.

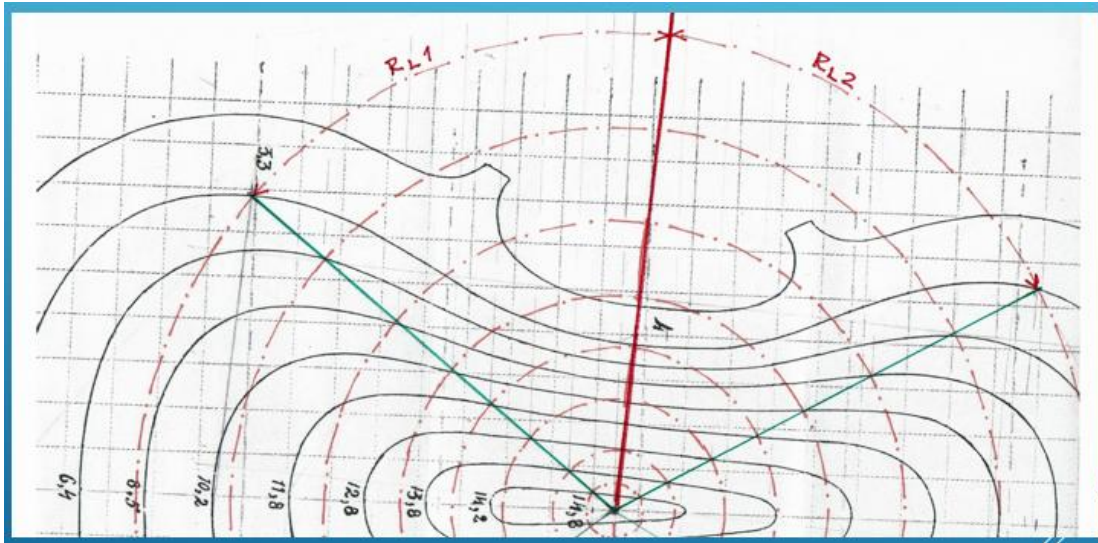


Figure 2.3

2.8 Sacconi's Search for Structural Logic

It is evident that Simone F. Sacconi was actively searching for a construction model—a reproducible framework capable of explaining the structural coherence of these master instruments.

Without access to computer-assisted design tools or digital modeling techniques, he relied instead on manual drafting and intuitive geometric reasoning.

My analysis of Sacconi's layout reveals a total of **sixteen cross-radii**, distributed as follows:

- **Six** in the upper bout area
- **Seven** in the lower bout
- **Three** in the C-bout

These radii form the basis for a geometrically consistent framework that supports both structural interpretation and design logic.

The figure below illustrates a **derived baseline**, constructed by extending the cross-arc shape at the height of the rib structure.

This baseline acts as a supporting framework for the entire arching system.

Each cross-sectional radial shape finds its **reactive counterpart**, accompanied by cross-arc shapes extending along the centerline of both the top and back plates.

The **red line**, designated as the base line, serves as a reference axis for geometric alignment.

The method for determining this baseline is shown below: the endpoints of the cross-radii are identified as **vault locations**—key structural points that define the curvature and load-bearing behavior of the plates.

To preserve geometric coherence and optimize structural equilibrium, the baseline itself must also be constructed using radii.

This ensures **continuity and symmetry** throughout the layout.

The resulting model became well-suited for both **physical construction** and **computational modeling** of the “iso-line” structure, offering a reproducible framework for violin design and analysis.

2.9 Baseline Point Calculation

The radii shapes were calculated from iso-line level heights by constructing chord lines and extending the radii to the 0,00 baseline point.

Through this process, **16 baseline points** were identified, see figure 2.4

Each of these points serves as a vault anchor:

- They mark the transition between successive radii.
- They stabilize the curvature sequence along the baseline.
- They provide the geometric foundation for iso-line construction and dipole analysis.

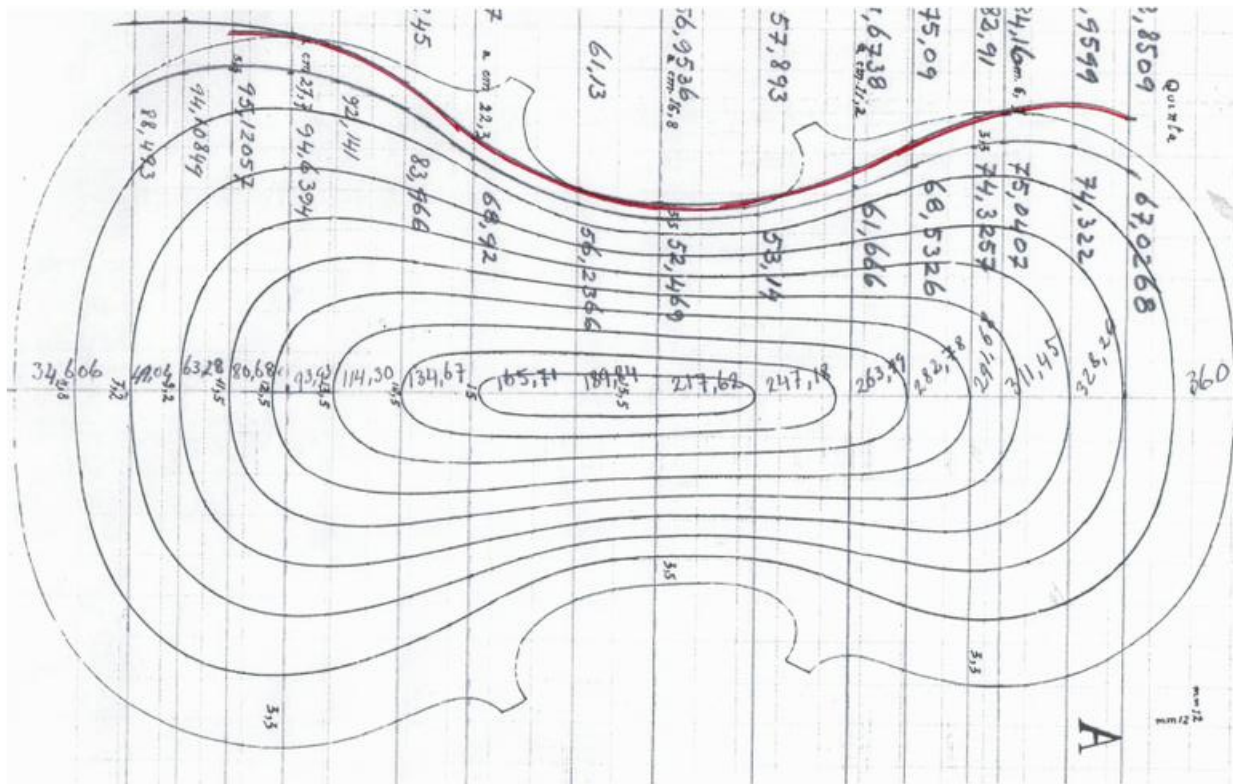


Figure 2.4

The Found Baseline (red) Based on Sacconi's Average Stradivari Iso-Line Layout

The baseline shown in red is derived from Sacconi's reconstruction of average Stradivari iso-lines.

While Sacconi's layout represents an empirical contour, the baseline here reveals the underlying geometric condition that can be reproduced and extended.

This alignment demonstrates how descriptive averages can be transformed into a constructive framework, bridging observation with geometric logic.

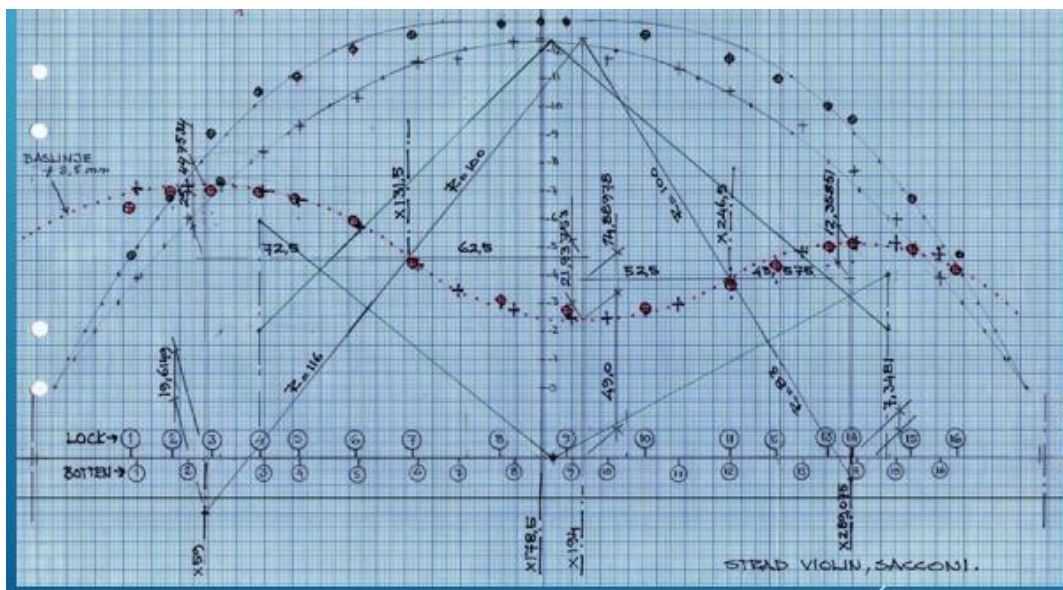


Figure 2.5

The baseline is not simply a descriptive contour but a **constructive element** see figure 2.5.

By defining radii and their center points, the layout becomes computationally tractable: iso-lines can be calculated systematically, ensuring accuracy and reproducibility.

This approach bridges **geometric intuition** with **numerical computation**, laying the groundwork for advanced structural analysis and later FEA investigations.

2.10 Visualizing the Baseline and Vault Geometry

Sacconi's contribution lay in his empirical reconstruction of Stradivari's arching.

Through iso-line drawings, he presented what he regarded as the average arching shape, supported by a minor constructive element in the central region.

We do not know how he arrived at these drawings; we only see the results.

Importantly, Sacconi **did not propose** any baseline model.

His work remained descriptive, offering contours but not a reproducible geometric framework.

The framework presented here builds upon Sacconi's observations across many Stradivari instruments, but seeks to translate them into a **reproducible drawing** that reveals the underlying structural logic.

By introducing the baseline and vault geometry, the arching can be described not as an empirical average but as a sequence of radii anchored on a geometric axis.

By calculating each radius from the chord length beneath its arc—at the height given—we can determine how that radius extends to the 0.00 reference level.

Each resulting point functions as a vault anchor, marking the transition between successive radii and stabilizing the arching system.

This reproducible baseline, when correctly composed, provides the geometric foundation for iso-line construction and enables the evaluation of dipole behaviour.

2.11 How the Geometric System Was Identified and Established

A useful geometric system has now been established.

This system makes it possible to adjust the baseline construction in order to search for arching shapes that support the dipole function. Initially, it seemed possible to construct the baseline from six radius shapes applied from the outset.

However, the resulting calculations did not produce equal cross-sections in the upper and lower bouts.

The assumption that the differing bout widths might determine the curvature of the baseline also failed to provide a satisfactory solution.

After extensive work—more than twenty attempts—the correct result finally emerged when the baseline curve was composed of only three equal radii connected at 60-degree angles.

This simply constructed baseline forms the core line of the arching system and is responsible for producing dipole behaviour. Figure 2.6 shows this outcome.

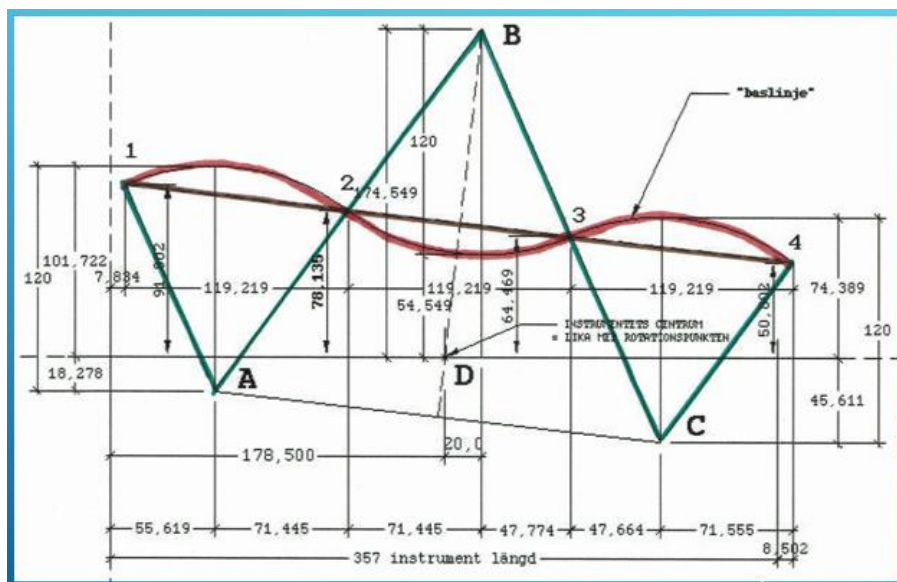


Figure 2.6, 2D “base Line” construction

2.12 The Found 2D Geometric Solution Producing Dipole Behaviour

The next task was to determine how the upper and lower bout structures could produce a reproducible dipole condition.

This required manipulating the size and location of the radii and their center points to test possible structural arrangements.

More than twenty different layouts were examined, each producing iso-line drawings calculated with computer assistance at 5 mm intervals along each level.

This was painstaking work, repeatedly tested without success, until a breakthrough occurred: a geometric layout constructed with **equal radii** arranged in a structural condition defined by **60-degree connections**.

This layout revealed the sought-after structural dipole condition.

The solution produced a beautiful and coherent transition from the upper arching shape to the lower arching shape, confirming that dipole behaviour can be embedded through **simple geometric construction** rather than empirical averaging.

Figure 2.1 shows that **thousands of equal cross-sections** would arise based on the “base line” and the length arc shape.

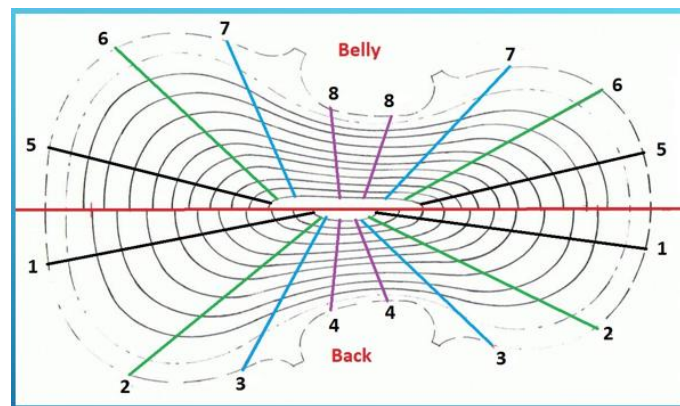


Figure 2.1, Result of 3D cross arc shapes on arching

This 2D geometric layout produces the important dipole condition.

Figure 2.7 illustrate the construction of the arching showing the six crosssection used by Stradivari

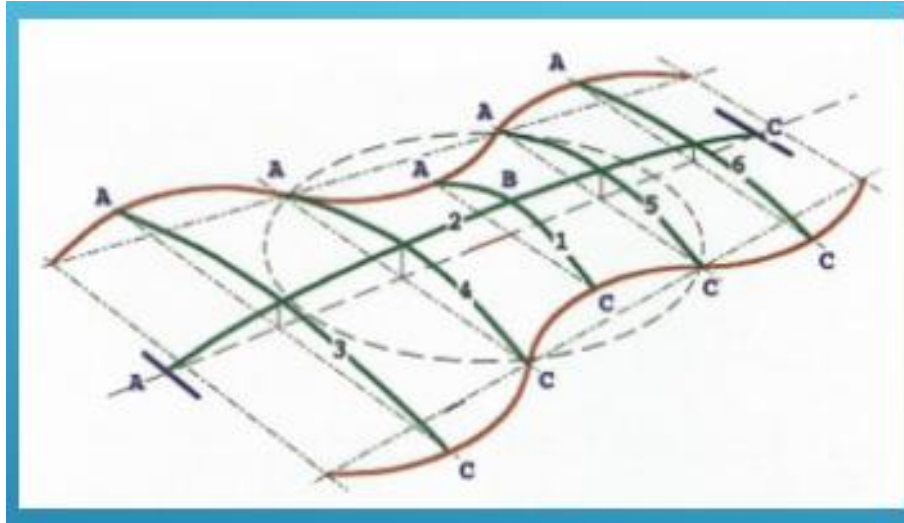


Figure 2.7, 3D cross arc construction

2.13 Bridging Geometry and Computation: Mapping the Iso-Line Structure

Using the geometric model, it became possible to compute precise **X, Y, and Z coordinates** at 5 mm intervals for any iso-line level, via a straightforward **BASIC program developed in 1983**.

This computational approach significantly enhanced mapping accuracy, enabling detailed visualization of the violin's structure through iso-lines.

Moreover, it provided a robust foundation for optimizing design parameters and investigating structural dynamics with both analytical clarity and geometric consistency.

The process also enabled the creation of **hand-drawn iso-lines**, which proved invaluable for analyzing the structural conditions necessary to achieve the desired outcome: the earlier-mentioned **dipole function**, expressed through equal cross-sections from the outline into the center.

2.14 Finalizing the Model: Symmetry, Adaptation, and Dynamic Function

The primary objective was to develop a model with a **distinctive, uniform cross-section** that transitions seamlessly from the instrument's outer outline to its central region—as illustrated in the figure above.

This design approach successfully integrates the **upper and lower bout shapes**, creating a harmonious and cohesive structure that supports the violin's constructive integrity.

In particular, it anticipates the **deflections resulting from string loading**, ensuring that the geometry is not only elegant but also dynamically functional.

By bridging geometric construction with computational mapping, the model achieves a reproducible dipole condition, integrating symmetry and dynamic adaptation into the violin's structural logic.

2.15 Möckel as a Bridge Figure

Otto Möckel (1869–1937), in *Die Kunst des Geigenbaues*, provided some of the earliest systematic iso-line measurements of classical violins, including instruments by J. Guarneri and Stradivari. See the figures 2.8 and 2.9.

His work was empirical: contours were traced and documented, offering valuable descriptive evidence of thickness distribution and arching form.

While Möckel did not propose a geometric construction method, his observations remain crucial.

They serve as a **historical bridge** between the intuitive craftsmanship of the old masters and the analytical reconstructions of modern research.

By aligning Möckel's empirical iso-lines with a reproducible geometric framework, this study demonstrates that the coherence he documented was not accidental.

Instead, it reflects an underlying **geometric logic** that classical luthiers may have employed instinctively.

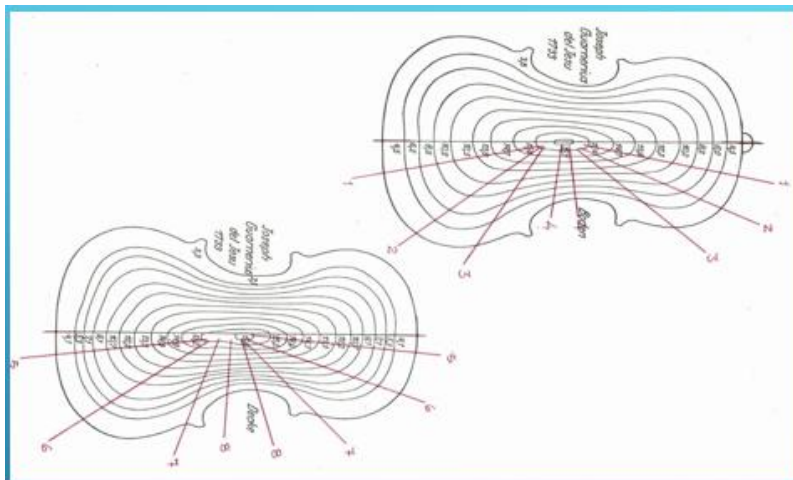


Figure 2.8

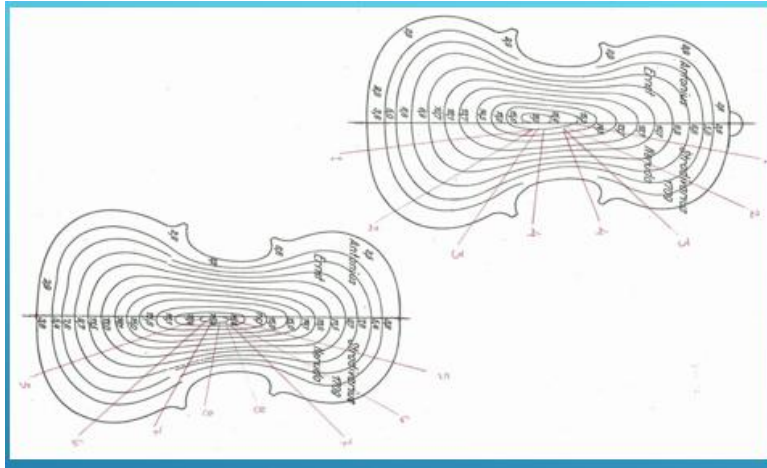


Figure 2.9

Iso-Line Observation by O. Möckel of J. Guarneri and Stradivari Violins

Empirical iso-line contours documented by O. Möckel, based on instruments by J. Guarneri and Stradivari.

The observed structural condition—with equal curve shapes and coherent transitions—reveals a pattern that aligns closely with the geometry-based model developed in this study. This correlation supports the hypothesis that classical luthiers may have used and understood, intuitively or deliberately, the geometric layout shown above, which produces structural coherence and the dipole quality on bout structures.

With Möckel's empirical iso-line observations now aligned to a reproducible geometric framework, the study advances to examine how these constructive conditions can be extended and refined within the broader logic of violin arching.

2.16 Comparative Note on Iso-Lines

It is important to note that **Sacconi's reconstructed iso-lines lack the presence of equal structural curve shapes**, reflecting descriptive averages rather than constructive coherence.

By contrast, **Möckel's empirically measured contours reveal these equal curve forms with clarity**, providing direct confirmation of the geometric coherence reconstructed in this study.

2.17 Geometry as Insight: From Square to Structure

This geometric layout solution encompasses both **2D and 3D representations**, offering a comprehensive framework for structural analysis and design.

At its core lies a **baseline structure** formed by three identical radii, seamlessly joined at 60-degree angles.

This configuration is as unexpected as it is refined—a product of geometric intuition that distills complexity into a harmonious structural core.

Tracing back to its origins, the baseline begins with a **square**.

Through triangular constructions, the layout emerges naturally, yielding a geometry that is both robust and elegant.

This breakthrough enabled the generation of thousands of uniform cross-sections, culminating in a model that is clear, efficient, and primed for advanced analysis.

Thanks to precise numerical calculations, the model has produced **high-fidelity data and invaluable insights**, forming the foundation for subsequent Finite Element Analysis (FEA) investigations.

The congruential consistency, though initially unanticipated, revealed itself as a striking affirmation of **geometric harmony's power** to shape structural integrity and functional behavior.

This discovery underscores how mathematical elegance can transcend abstraction, becoming a tangible force in design innovation.

Doubling the figure reveals the initial shape, which aligns seamlessly with the final layout—underscoring the elegance and effectiveness of this geometric approach.

It highlights the foundational role of geometry in achieving **structural coherence and dynamic functional design**.

What is especially remarkable is that a master of geometric construction, working without modern computational tools, conceived such a precise and functionally optimized instrument form.

It stands as a testament to **intuitive engineering and profound structural awareness**—where insight and craftsmanship converge to transcend technological limitations.

From the square to the structure, geometry emerges not as ornament but as the **hidden principle of coherence, bridging intuition and analysis across centuries of violin making**.

See figures 2.10 and 2.11

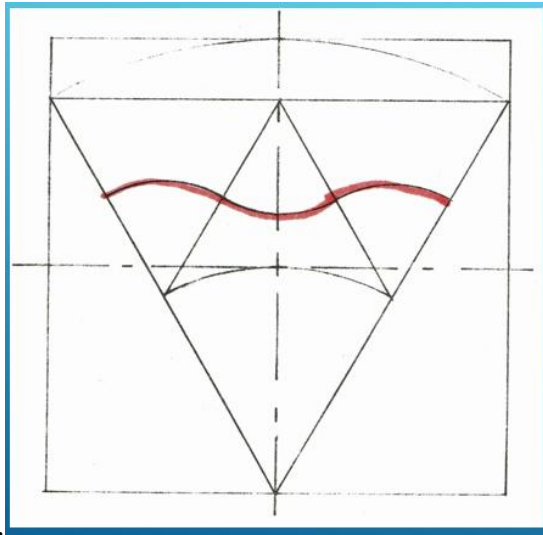


Figure 2.10

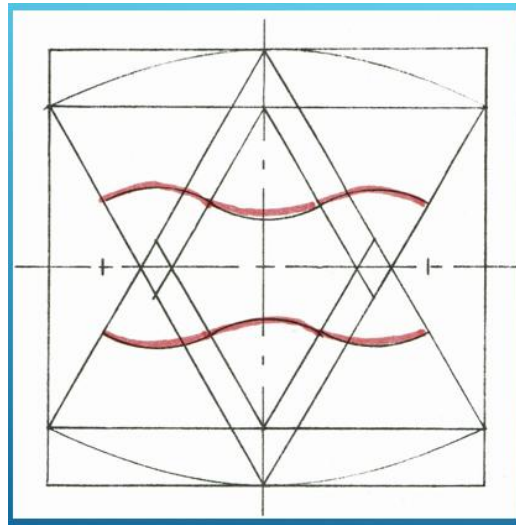


Figure 2.11

2.18 Baseline and Rotational Transformation of the Geometric Layout

The baseline, defined by radii and center points, establishes the constructive foundation for iso-line calculations.

When rotated clockwise and counterclockwise, the figure reveals the violin's distinctive structural identity: symmetry clarifies the positioning of C-corners, F-holes, fingerboard taper, and bass bar alignment, while the longitudinal axis remains unaffected.

Together, the baseline and its rotational transformation demonstrate how simple geometric principles evolve into a coherent framework that governs both structure and function.

2.19 Collision at Point “a” and Structural Relations

The rotational geometry of the two figures reaches its limit when they collide at **point “a.”**

At this moment, the layout reveals a simple yet comprehensive explanation of the instrument's special structural relations.

The collision clarifies how internal features—such as the positioning of the **C-corners**, the orientation and length of the **F-holes**, the location, the length of the **strings**, the tapering of

the **fingerboard**, and the alignment of the **bass bar**—are all coherently integrated within the geometric framework.

Most important, however, is the construction of the **arching shape**, whose special quality produces the **dipole layout**.

This stopping point demonstrates that the geometry is not arbitrary: the collision at point “a” functions as a **structural key**, exposing the hidden logic that governs both the instrument’s form and its dynamic behavior.

The figure 2.12 to 2.15

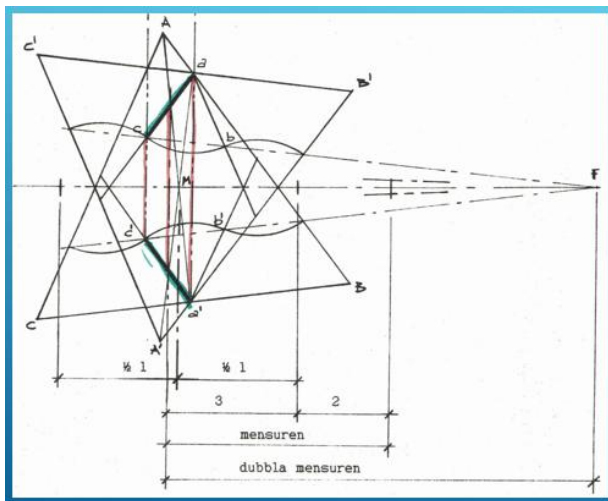


Figure 2.12

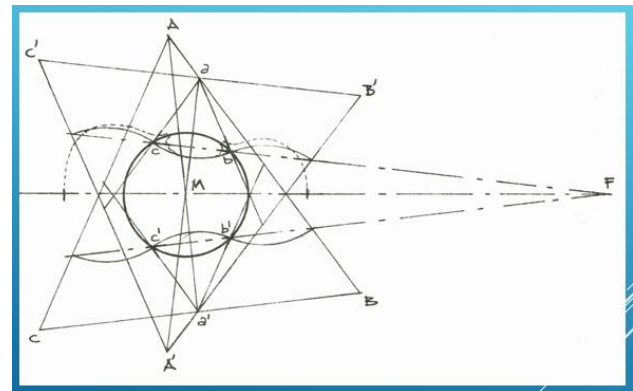


Figure 2.13

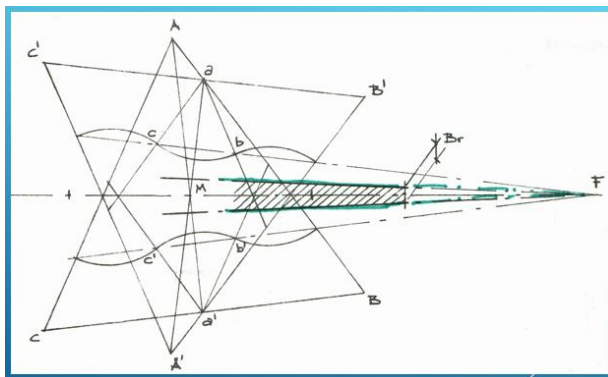


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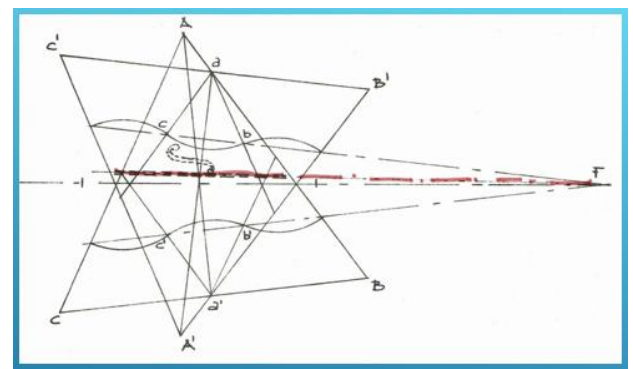


Figure 2.15

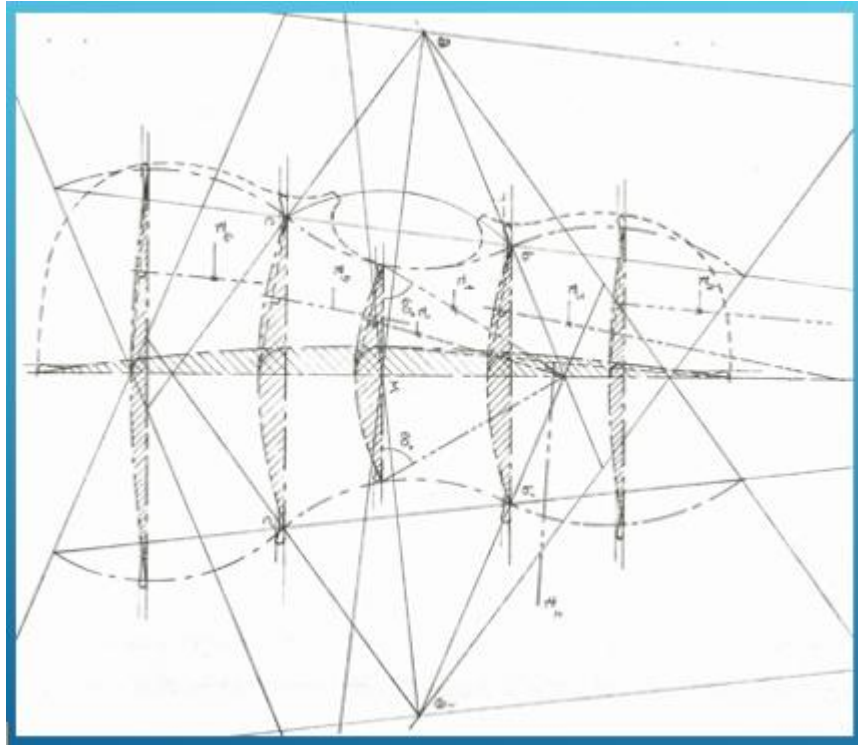


Figure 2.16

2.20 The Six Arching Templates of Stradivari

Among the artifacts preserved in Cremona are the **six wooden templates** attributed to Antonio Stradivari. See Figure 2.16.

The locations of these templates, shown above, indicate their role in shaping the **arching of the plates**, providing a reproducible guide for the curvature of the violin's top and back.

The figure demonstrates where these template shapes originate: they are not arbitrary, but emerge directly from the **geometric framework reconstructed in this study**.

By aligning the templates with the derived construction lines, their coherence becomes evident.

Each template reflects a segment of the arching logic, confirming that Stradivari's practice was grounded in a systematic geometric approach rather than empirical averaging.

Thus, the "six" serve as tangible evidence of the **structural identity** embedded in the Cremonese tradition: geometry translated into craft, preserved in wood, and transmitted through generations of luthiers.

2.21 Toward Structural Tuning: A Preview of Chapter 3

As will be shown in Chapter 3, there exists another **critical structural quality** that significantly influences the function of the instrument.

I have long searched for this structural quality, and its identification marks a decisive turning point in the study.

Once recognized within the demonstrated geometry, it became clear how a luthier can improve sound quality by **individually tuning the bout structures**, thereby altering the instrument's behavior in ways that markedly enhance tonal output.

This discovery suggests that the **upper and lower bouts** can be treated as distinct acoustic zones in relation to an intern frame work, each zone capable of adjustment to influence the instrument's overall response and ultimately produce **constructive interference**.

The old master luthiers may well have understood this possibility—certainly not through formal analysis, but through **intuitive craftsmanship and acoustic sensitivity**.

Their insight into the responsiveness of the bout structures points to a tradition where **geometry and tuning converge**, shaping both structural integrity and tonal excellence.

2.22 The Instrument's Outline Structure Explained

The instrument's outline is directly related to the **arching shape**. To examine this, we consider the behavior of the **iso-lines**.

Sequential rotation of each iso-line to align with the upper plane layout reveals:

- The **lowest iso-line** requires significant rotation.
- The **highest iso-line** requires almost none.

Applying the same rotational method to the instrument's outline—as seen in the Stradivari example—shows that the lower, wider section is geometrically related to the upper, closely resembling the calculated iso-lines derived from the reconstructed geometry.

- **Blue lines:** original iso-lines from the geometric model, mainly covered.
- **Red lines:** the same iso-lines, rotated and mirrored to align covering the blue.
- **The matching curvatures confirm geometric coherence**, demonstrating that the outline follows the same structural logic as the arching iso lines.

Importantly, the iso lines do not maintain a constant radius, and therefore neither the instrument's outline nor the outline of the mould on which the ribs are constructed can be based on fixed-radius curves.

The congruence between the red and blue lines demonstrates the instrument's structural coherence and confirms that the outline is derived from the iso-line arching geometry, rather than the reverse.

The turnover line configuration, based on the described geometry, shows that the outline emerges from the arching shape's iso-line.

Thus, the instrument's form is generated by the arching system, not imposed upon it. This relationship is illustrated in Figures 2.17 and 2.18.



Figure 2.17

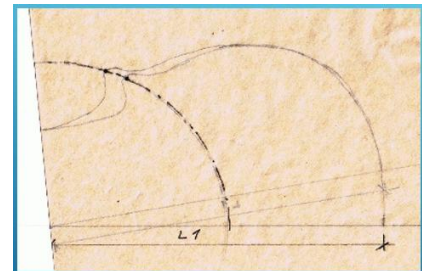


Figure 2.18

Figures 2.17 and 2.18.

Rotation of the iso-lines is marked at reference points A–F1.

These markers trace the congruence between the red and blue paths, demonstrating that the instrument's outline is generated by the rotational geometry of the arching iso-lines.

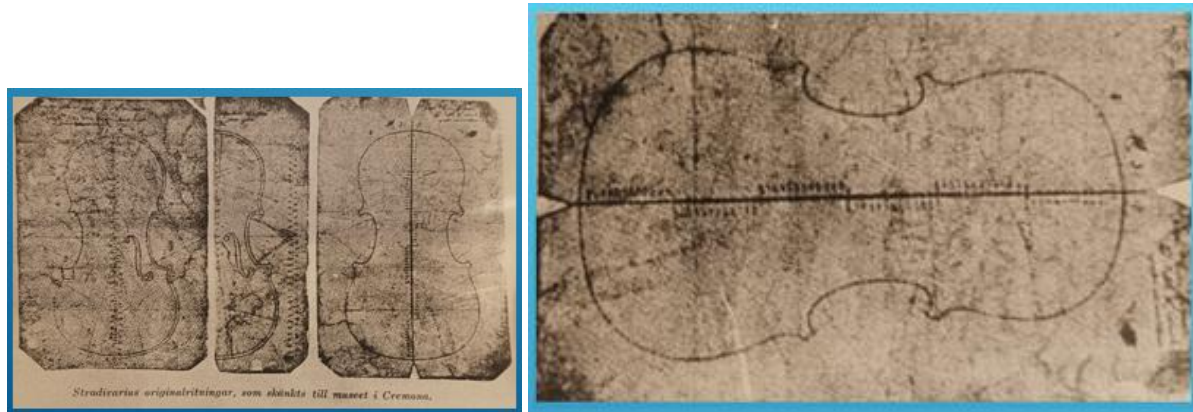
The configurational perimeter therefore arises from internal structural logic rather than serving as a generative boundary.

From the internet we encounter the following image, although its original source is unknown.

It is presented as a set of drawings attributed to Stradivari.

Notably, the outline configuration displays the same conditions shown in Figures 2.17 and 2.18, where the lower-bout outline folds over and aligns with the upper-bout outline.

The bout corners lie on the same circle, with its centre positioned at the midpoint of the instrument's length.



We learn that the outline curve of the lower bout contains the same curve that appears in the upper bout.

Geometric Alignment with Master Instruments

This principle establishes a direct geometric correspondence with master instruments, in which the outline consistently emerges as a consequence of the arching iso-line system—not the other way around.

Figure 2.17 confirms this relationship. In this way, the perimeter reflects inherited structural coherence rather than an imposed design.

This congruence demonstrates that the discovered geometry is not merely theoretical but structurally embedded in historical craftsmanship, grounded in simple 2D–3D geometric relationships.

What emerges is a unified structural language—one that validates the intuitive engineering of the old masters and affirms the enduring relevance of geometry-based design.

2.23 Structural 2D–3D Layout Confirmation

Figures 2.19 and 2.20. The discovered 2D structural layout fits precisely onto photographs of master instruments by Stradivari and Guarneri del Gesù.

This congruence confirms that the geometry arises from embedded 3D structural logic, not surface design, and reflects principles deeply rooted in historical craftsmanship.

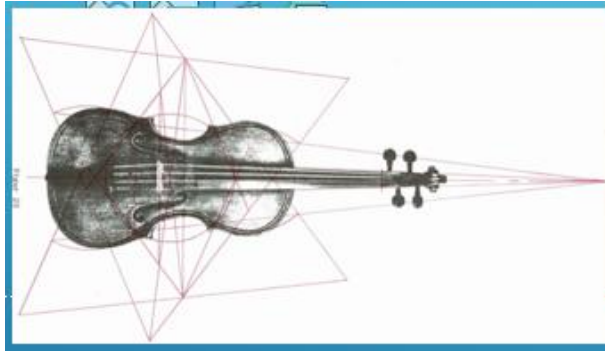


Figure 2.19 Guarneri del Gesù “Wieniavski

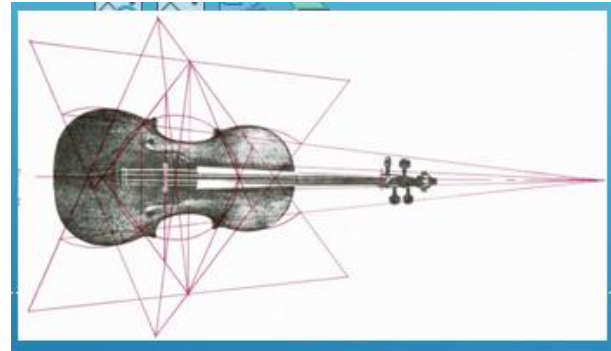


Figure 2.20 Stradivarius “Harrison

2.24 Thus far explained.

In this pivotal chapter, a previously hidden geometric framework reveals its full significance.

What had seemed a collection of isolated observations now coalesces into a coherent structural logic—a breakthrough that bridges constructional design with the geometry introduced earlier.

This marks the transition from geometric intuition to functional understanding, a shift that empowers the luthier to actively control mechanical and dynamic behavior, and thereby refine acoustic output.

Geometry is no longer a passive reference; it becomes an active mechanism for shaping the instrument’s vibrational response, modal balance, and tonal clarity.

2.25 Mechanical Control and Responsive Behavior

To advance from geometric insight to functional mastery, we must address the realm of mechanically controllable behaviors. These are the structural responses that can be intentionally shaped through design—behaviors that govern how an instrument flexes, vibrates, and transmits energy under playing conditions.

This includes:

- **Localized deformation zones** that respond predictably to string loading, providing controllable points of adjustment.

- **Spring-like behavior** distributed across the arching contours, producing stress in specific, controllable areas.
- **Dipole dynamics** emerging from coordinated bout interaction, yielding a controllable “breathing” quality.

By identifying and refining these mechanical control points, the luthier gains the ability to modulate acoustic output, enhance tonal clarity, and optimize structural equilibrium. Geometry thus becomes the tool not only for shaping form, but for actively controlling function.

Conclusion of Chapter 2

This chapter has demonstrated how a simple geometric construction reveals the conditions under which dipole qualities emerge. By shaping the baseline in relation to the fixed-length arc, the violin’s form achieves congruence between upper and lower bouts, thereby defining the outline of the instrument.

These conditions not only produce a breathing dynamic within the corpus but also give rise to constructive interference, strengthening the instrument’s internal coherence. Yet these qualities do not exist in isolation. They are closely connected to a deeper structural framework that governs the stability of the arching itself.

This framework will be presented in Chapter 3, where the arching system and its internal geometry are shown to stabilize the arching shape and extend the principles introduced here into a comprehensive structural logic.

In turn, **Chapter 3** will demonstrate how this stabilizing framework directly influences dipole quality and constructive interference, enabling the luthier to harness the instrument’s functions effectively in practice.